

Tutorial Sheet -05
Classful and Classless IP Addressing and
Subnetting

Course: B.Tech Hons.

Subject: Data Communication and Network System

Name of the faculty: Ms. Chhavi Bajpai

Section: EA, EB

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Q1. A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet? **(GATE 2004)**

- A. 14
- B. 30
- C. 62
- D. 126

Q2. The routing table of a router is shown below: **(GATE 2004)**

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
Default		Eth2

On which interface will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively?

- A. Eth1 and Eth2
- B. Eth0 and Eth2
- C. Eth0 and Eth3
- D. Eth1 and Eth3

Q3. An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be: **(GATE 2005)**

- A. 255.255.0.0
- B. 255.255.64.0
- C. 255.255.128.0
- D. 255.255.252.0

Q4. A company has a class C network address of 204.204.204.0. It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs? **(GATE 2005)**

- A. 204.204.204.128/255.255.255.192
204.204.204.0/255.255.255.128
204.204.204.64/255.255.255.128
- B. 204.204.204.0/255.255.255.192
204.204.204.192/255.255.255.128
204.204.204.64/255.255.255.128
- C. 204.204.204.128/255.255.255.128
204.204.204.192/255.255.255.192
204.204.204.224/255.255.255.192

- D. 204.204.204.128/255.255.255.128
204.204.204.64/255.255.255.192

Q5. A router uses the following routing table:

(GATE 2006)

Destination	Mask	Interface
144.16.0.0	255.255.0.0	eth0
144.16.64.0	255.255.224.0	eth1
144.16.68.0	255.255.255.0	eth2
144.16.68.64	255.255.255.224	eth3

Packet bearing a destination address 144.16.68.117 arrives at the router. On which interface will it be forwarded?

Q6. Two computers C1 and C2 are configured as follows:

C1 has IP address 203.197.2.53 and netmask 255.255.128.0.

C2 has IP address 203.197.75.201 and netmask 255.255.192.0. Which one of the following statements is true?

(GATE 2006)

- A. C1 and C2 both assume they are on the same network
- B. C2 assumes C1 is on same network, but C1 assumes C2 is on a different network
- C. C1 assumes C2 is on same network, but C2 assumes C1 is on a different network
- D. C1 and C2 both assume they are on different networks.

Q7. If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(GATE 2008)

Q8. Host X has IP address 192.168.1.97 and is connected through two routers R1 and R2 to another host Y with IP address 192.168.1.80.

Router R1 has IP addresses 192.168.1.135 and 192.168.1.110. R2 has IP addresses 192.168.1.67 and 192.168.1.155. The netmask used in the network is 255.255.255.224.

Which IP address should X configure its gateway as?

(GATE 2008)

- A. 192.168.1.67
- B. 192.168.1.110
- C. 192.168.1.135
- D. 192.168.1.155

Q9. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

(GATE 2012)

- A. 245.248.136.0/21 and 245.248.128.0/22
- B. 245.248.128.0/21 and 245.248.128.0/22
- C. 245.248.132.0/22 and 245.248.132.0/21
- D. 245.248.136.0/24 and 245.248.132.0/21

Q10. Suppose computers A and B have IP addresses 10.105.1.113 and 10.105.1.91

respectively and they both use same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network? **(GATE 2012)**

- A. 255.255.255.0
- B. 255.255.255.128
- C. 255.255.255.192
- D. 255.255.255.224

Q11. An IP router implementing Classless Inter-domain Routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries:

(GATE 2014)

Prefix	Outer Interface Identifier
131.16.0.0/12	3
131.28.0.0/14	5
131.19.0.0/16	2
131.22.0.0/15	1

The identifier of the output interface on which this packet will be forwarded is ____.

Q12. Consider the following routing table at an IP router:

(GATE 2015)

Network No	Net Mask	Next Hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
0.0.0.0	Default	R4

For each IP address in Group I Identify the correct choice of the next hop from Group II using the entries from the routing table above.

Group I	Group II
i) 128.96.171.92	a) Interface 0
ii) 128.96.167.151	b) Interface 1
iii) 128.96.163.151	c) R2
iv) 128.96.164.121	d) R3
	e) R4

- A. i-a, ii-c, iii-e, iv-d
- B. i-a, ii-d, iii-b, iv-e
- C. i-b, ii-c, iii-d, iv-e
- D. i-b, ii-c, iii-e, iv-d

Q13. Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true? **(GATE 2019)**

- A. M, N, and P all belong to the same subnet
- B. Only M and N belong to the same subnet
- C. Only N and P belong to the same subnet
- D. M, N, and P belong to three different subnets

Q14. Consider routing table of an organization's router shown below: **(GATE 2022)**

Subnet Number	Subnet Mask	Next Hop
12.20.164.0	255.255.252.0	R1
12.20.170.0	255.255.254.0	R2
12.20.168.0	255.255.254.0	Interface 0
12.20.166.0	255.255.254.0	Interface 1
default		R3

Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

- A. 12.20.164.0/20
- B. 12.20.164.0/22
- C. 12.20.164.0/21
- D. 12.20.168.0/22

Q15. The forwarding table of a router is shown below. **(GATE 2023)**

Subnet Number	Subnet Mask	Interface ID
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.224	4
Default		0

A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID _____.

Q16. Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255? **(GATE 2024)**

- A. 10.12.2.0/23
- B. 10.12.2.0/24
- C. 10.12.0.0/22
- D. 10.12.2.0/22

Q17. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28.

- (i) What is the first address in the block?
- (ii) What is the last address in the block?
- (iii) How many addresses are in the block?

Q18. The only entries in a certain route table are (128.59.28.0/22, port 0), (128.59.28.0/23, port 1) and (128.59.28.0/24, port 2). These entries indicate CIDR network number, the prefix indication and the corresponding port to which a packet should be forwarded. If a packet arrives with a destination IP address equal to 128.59.29.18, which port will this router forward the packet to?

Q19. Suppose an ISP owns the block of addresses of the form 131.210.16.0/21. Suppose it wants to create four subnets from this block, with each block having the same number of IP addresses. **(i)** What are the prefixes (of form a.b.c.d/x) for the four subnets? **(ii)** What is the number of host address that can be supported per subnet?

Q20. How many bits are allocated for network id (NID) and host id (HID) in the IP address 25.193.155.233?

- A. 24 bit for NID, 8 bits for HID
- B. 8 bit for NID, 24 bits for HID
- C. 16 bit for NID, 16 bits for HID
- D. none